SPARKY'S SCIENCE EXPERIMENTS

Paper Aeroplane: The Forces of Flight

Physical Science (forces) · Ages 7+ · 30 minutes

Fold a dart paper plane (paper aeroplane), test how a paper clip on the nose and wing flaps change the way it flies, and fix a plane that dives or drops.

Safety first

Sparky always says: Never throw your plane at people or animals - the point can hurt an eye!

- Throw it in an open space.
- Ask an adult to help with scissors.

You need

- A sheet of A4 paper (scrap paper is fine)
- Paper clips and scissors
- A tape measure, or count your steps
- An open space like a school field

Steps

1. Fold the paper in half the long way and open it. Fold the two top corners to the middle line.
2. Fold the new slanted edges to the middle line again.
3. Fold the whole thing in half along the middle line.
4. Fold each wing down to line up with the bottom of the body. Throw it gently, straight ahead.
5. Test 1: throw it 3 times and measure. Add a paper clip to the nose and throw 3 more times.
6. Test 2: cut two slits 1 cm apart at the back of each wing and bend the flaps up a little. Then try them bent down.

If it doesn't work, try these in order

1. Dives into the ground? Bend the back flaps up a little. Still dives? Take off any nose paper clip and bend the flaps up more.
2. Climbs up, then drops? Flatten the flaps.
3. Turns to one side? Make both wings the same shape and size.

Why it works

Four forces act on a plane. Thrust is your throw pushing it forward. Drag is the air pushing back. Weight is gravity pulling down. Lift is the air pushing up: the wings push air down, so the air pushes the wings up. A paper clip on the nose helps it fly straight. Flaps bent up lift the nose for more lift - but too much and the plane slows and drops.

Did you know? Planes at O.R. Tambo Airport in Johannesburg need a longer run to take off than at the coast, because the air is thinner high up on the Highveld.

Ask Sparky: Why do real planes have flaps on their wings?